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Spoken Languages: English

EDUCATION

May 2020 – May 2024	Boston University, Boston, MA, USA Doctor of Philosophy, Astronomy Dissertation Title: Properties of Voids and Void Galaxies Research Advisor: Tereasa G. Brainerd, PhD
September 2018 – May 2020	Boston University, Boston, MA, USA Master of Arts, Astronomy Graduate Coursework: High-energy Astrophysics, Galaxy Evolution, Radiative Transfer, Plasma Physics, Fluid Mechanics
August 2014 – May 2018	University of South Florida, Tampa, FL, USA Bachelor of Science, Physics Bachelor of Arts, Mathematics (pure concentration) USF Honors College Summa Cum Laude Graduate Coursework: Computational Physics, General Relativity

RESEARCH PUBLICATIONS & PAPERS

- [1] **Curtis, O.**, McDonough, B., and Brainerd, T. G. "Void Galaxies and AGN Activity in ZOBOV-Identified TNG300 Voids out to $z=3.0$." (In preparation.)
- [2] McDonough, B., **Curtis, O.**, and Brainerd, T.G. "Intrinsic and Environmental Effects on the Distribution of Star Formation in TNG100 Galaxies." (Under review.)
- [3] **Curtis, O.**, McDonough, B., and Brainerd, T. G. "Evolution of Density Profiles and Bias Estimates in ZOBOV-Identified Voids in the TNG300 Simulation." (Under review.)
- [4] **Curtis, O.**, McDonough, B., and Brainerd, T. G. "Properties of Voids and Void Galaxies in the TNG300 Simulation." *The Astrophysical Journal* 962.1 (2024): 58. doi: 10.3847/1538-4357/ad18b4
- [5] McDonough, B., **Curtis, O.**, and Brainerd, T. G. "Resolved Star Formation in TNG100 Central and Satellite Galaxies." *The Astrophysical Journal* 958.1 (2023): 19. doi: 10.3847/1538-4357/acfe0b
- [6] **Curtis, O.**, Brainerd, T. G., and Hernandez, A. "Cosmic voids in GAN-generated maps of large-scale structure." *Astronomy and Computing* 38 (2022): 100525. doi: 10.1016/j.ascom.2021.100525
- [7] Womack, M., **Curtis, O.**, et al. "The Visual Lightcurve of Comet C/1995 O1 (Hale–Bopp) from 1995 to 1999." *The Planetary Science Journal* 2.1 (2021): 17. doi: 10.3847/PSJ/abd32c

- a. Software Release: **Curtis O.**, et al. "ICQSplitter: Python Tool for Handling Visual or CCD Magnitudes of Comets v3.0," *Zenodo*, (2020) doi: 10.5281/zenodo.3628044
- [8] **Curtis, O.** and Brainerd, T. G. "Fast generation of large-scale structure density maps via generative adversarial networks." *Research Notes of the AAS* 4.6 (2020): 90. doi: 10.3847/2515-5172/ab9e01
- [9] Chandler, C. O., **Curtis, O.**, et al. "SAFARI: Searching asteroids for activity revealing indicators." *Publications of the Astronomical Society of the Pacific* 130.993 (2018): 114502. doi: 10.1088/1538-3873/aad03d
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INVITED TALKS

- [1] **Curtis, O.**, et al. "Voids and Void Galaxies in the TNG300 Simulation." The Co-evolution of the Cosmic Web and Galaxies Across Cosmic Time, The Kavli Institute for Theoretical Physics, Santa Barbara, California, February 6th – 9th, 2023. Oral Presentation.
- [2] **Curtis, O.**, et al. "Voids and Void Galaxies in the TNG300 Simulation." Physics and Astronomy Seminar Series, Tufts University, October 22nd, 2021. Oral Presentation.
- [3] **Curtis, O.** and Brainerd, T. G. "Fast Generation of Large-Scale Structure Density Maps via Generative Adversarial Networks." 236th Meeting of the American Astronomical Society, Virtual Conference, June 1st – 3rd, 2020. Interactive Poster Presentation.
- [4] **Curtis, O.**, et al. "SAFARI: Searching Asteroids for Activity Revealing Indicators." 232nd Meeting of the American Astronomical Society, Denver, Colorado, June 3rd – 7th, 2018. Interactive Poster Presentation.
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CONTRIBUTED TALKS

- [1] McDonough, B., **Curtis, O.**, et al. "Exploring IllustrisTNG simulations to derive observationally comparable star formation rates and metallicities." Recipes to Regulate Star Formation at All Scales: From the Nearby Universe to the First Galaxies, Baltimore, Maryland, April 15th – 19th, 2024. Workshop.
- [2] Lastra, N., **Curtis, O.**, et al. "Best Secular Lightcurve for Comet Hale-Bopp and Implications for Dust Production Rates." American Physical Society March Meeting, Los Angeles, California, March 5th – 9th, 2018. Poster Presentation.
- [3] Womack, M., Lastra, N., **Curtis, O.**, et al. "The Secular Lightcurve and Photometric Evolution of C/1995 O1 (Hale-Bopp)." 49th American Astronomical Society Annual Division of Planetary Science Meeting, Provo, Utah, October 15th-20th, 2017. Interactive Poster Presentation.
- [4] Harrington, O., Womack, M., **Curtis, O.**, et al. "Correlation Between Cometary Gas/Dust Ratios and Heliocentric Distance." 49th American Astronomical Society Annual Division of Planetary Science Meeting, Provo, Utah, October 15th-20th, 2017. Poster Presentation.
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GRANTS

- [1] 2023 BU Graduate Student Organization Travel Grant
 - [2] 2017 Society of Physics Students National Travel Grant
 - [3] 2017 USF Travel Grant
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AWARDS AND ACCOMPLISHMENTS

- [1] 2018 AAS Chambliss Astronomy Student Achievement Award
 - [2] 2017 Barry M. Goldwater Scholar Honorable Mention
 - [3] 2017 National Science Foundation Research Experience for Undergraduates: Northern Arizona University
 - [4] 2017 USF Undergraduate Research Conference's Excellence in Research Award
 - [5] USF Undergraduate Research Scholar
 - [6] USF Honors Scholarship
 - [7] Dean's List of Scholars: Fall 2014 – Spring 2018.
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TECHNICAL SKILLS

Software Languages: Python, Fortran, CUDA, C, MATLAB, Bash/Unix shell-scripting, R, SQL.

Frameworks & Libraries: NumPy, Astropy, GADGET, CONCEPT, pandas, scikit-learn, TensorFlow, PyTorch, PySQL, emcee, Matplotlib and seaborn for data visualization.

Developer Tools: Jupyter, Git, Docker, Slurm Workload Manager for high-performance computing

Methods: Integral Field Unit (IFU) spectroscopy, N-body simulations, MHD modeling, adaptive mesh refinement, deep and statistical learning, Bayesian Analysis, and MCMC methods. Comfortable navigating and extracting information from databases such as the Gaia Data Releases or the Sloan Digital Sky Survey, the Panoramic Survey Telescope and Rapid Response System.

Image Processing: IRAF/PyRAF, SAOImage/ds9, Source Extractor, SCAMP, and SWarp for handling of FITS images and generating astrometry and photometry of targets.

TEACHING EXPERIENCE

Instructor of Record

- [1] Boston University – CC 111 Core Natural Science I – Fall 2022
- [2] Boston University – CC 212 Core Natural Science II – Spring 2023

Teaching Assistant

- [1] Boston University – AS 101 The Solar System – Fall 2018, Spring 2019, Fall 2019, Spring 2019, Spring 2021, Summer 2021, Summer 2023
 - [2] Boston University – AS 100 Cosmic Controversies – Spring 2020
 - [3] Boston University – AS 102 The Astronomical Universe – Summer 2020, Summer 2021
 - [4] Boston University – AS 105 Alien Worlds – Fall 2020
 - [5] Boston University – AS 202 Principles of Astronomy I – Fall 2021
 - [6] Boston University – AS 203 Principles of Astronomy II – Spring 2022
 - [7] Boston University – AS 107 Life Beyond Earth – Spring 2024
 - [8] University of South Florida – AST 4721L Astrophysics Laboratory – Spring 2017
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PUBLIC OUTREACH

- [1] **Graduate Student Representative** – For the 2021-2022 academic year, I was a graduate student representative for the astronomy department at BU. I liaised between the faculty and my fellow graduate students, aiding in faculty hiring searches and graduate student admission processes. I also led weekly journal clubs, including a special weekly journal club in Spring 2022 that focused on physics education research and how to incorporate ideas into our own classrooms.
- [2] **Skype a Scientist** – Since Spring 2019, I have participated in the Skype a Scientist program. I have talked with over 20 K-12 classrooms across the country. There, I gave mini-lessons on space-based topics and talked to them about life as an astronomer.
- [3] **Public Observing Nights** – I volunteer at BU’s Judson B. Coit Observatory’s public open night about 15 clear nights a year. I help set up telescopes and answer any space-based questions guests have.
- [4] **Boston Waterworks Museum** – Once a month since 2021 I have volunteered at Boston’s Public Waterworks Museum. I’ve led tours, showed off the 150-year-old steam engines on display, and have developed interactive activities for our children’s exhibit.
- [5] **USF Engineering EXPO** – Led USF’s Society of Physics Students to perform multiple 45-minute shows to thousands of Tampa Bay public school students at USF’s annual Engineering EXPO. Years performed: 2016, 2017, and 2018.

REFERENCES

Professor Tereasa Brainerd	Professor Alan Marscher	Professor Paul Withers
Boston University	Boston University	Boston University
Primary Advisor, Dissertation	Course instructor, Dissertation	Committee Chair, I was a TA for him for 3 semesters
1 st reader	2 nd reader	
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Additional references are available upon request.